Fingerprint sensor security revisited: A cooperative and a non-cooperative method to bypass capacitive and optical sensors with almost perfect success ratio

Ryan Carvalho
xen1thLabs, Hardware Laboratory
Digital14 LLC
Abu Dhabi, UAE
ryan.carvalho@digital14.com

Norbert Tihanyi
xen1thLabs, Cryptography Laboratory
Digital14 LLC
Abu Dhabi, UAE
norbert.tihanyi@digital14.com

***Abstract*—In this paper we present both a cooperative and a non-cooperative direct attack method to bypass fingerprint sensors (optical scanners, capacitive scanners) used in embedded devices. Using the cooperative method a 96% success rate can be achieved using a combination of Ethylene-Vinyl Acetate –$(C_2H_4)_n(C_4H_6O_2)_m$– commonly known as "hot glue" and Polyvinyl Acetate –$(C_4H_6O_2)_n$– commonly known as "white glue" or "school glue". The cooperative method takes 5-10 minutes and costs no more than $5 which makes it very lucrative and accessible to anyone. We were able to achieve these high success rates on the latest hardware such as the Windows Hello Framework or certified high security devices. Using the non-cooperative method we were able to achieve 98% success ratio on optical sensors. During the experiment the following devices were tested: Suprema Biostation 2, 2019 Macbook Air, 2017 MacBook Pro, HUAWEI P9, HUAWEI P20, Lexar F35 encrypted USB drive, ASUS Zenbook UX461F Z14, iPhone 8. The whole process is precisely documented and disclosed in this paper for the research community.**

***Index Terms*—Fingerprint sensors, biometric security, hardware security bypass**

I. Introduction

In computer security there are three different ways to identify yourself, based on "what you know," such as passwords and PIN codes, "what you have" like certificates and hardware tokens, and "who you are" which is known as biometric authentication [1]. While the most widely used mechanism for user authentication is passwords [2], biometric security systems such as fingerprint authentication can be considered emerging technologies. Biometric techniques are being used increasingly especially in mobile hardware devices and laptops. The primary purpose of using a biometric system is to provide non-repudiable authentication [3]. Using biometric authentication systems an individual can be authenticated based on "who they are" rather than "what they know". As the technology is getting more popular, more and more attack vectors can be identified against biometric sensors. There are two major types of attacks [4] against biometric systems called *direct attacks* and *indirect attacks* as can be seen in Figure 1.

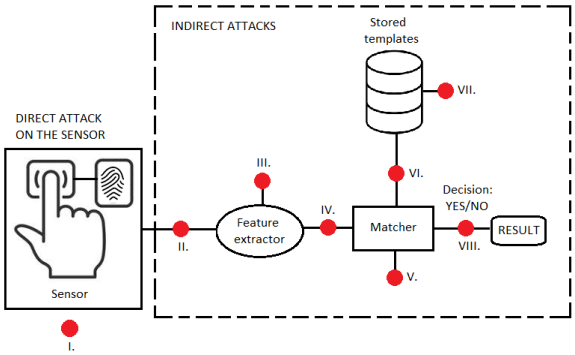

Fig. 1: Attack vectors against biometric systems

The direct attack is targeting the sensor itself, while indirect attacks are different types of attacks in different layers of the biometric system. Ratha [5] identified and classified eight attack points in a generic biometric system:

- **Type I - Sensor attack:** In this type of attack, a possible reproduction of the biometric data will be presented to the system. (e.g: fake finger)
- **Type II - Replay attack:** In this type of attack a previously recorded biometric data is replayed to the system bypassing the sensor.
- **Type III - Feature extract override attack:** In this type of attack the feature extractor could be attacked with a Trojan horse so that it would produce feature sets chosen by the attacker.
- **Type IV - Feature representation tampering:** After the features have been extracted from the input signal they are replaced with a different feature set (assuming the representation is known).
- **Type V - Override matcher:** The matcher is attacked to always directly produce an artificially high or low match score).
- **Type VI - Channel attack:** In this type of attack the

templates from the stored database are sent to the matcher through a channel which could be attacked to change the contents of the templates before they reach the matcher.
- **Type VII - Manipulate stored templates:** In this type of attack an attacker tries to modify the stored templates in the database which could result in authorization for a fraudulent individual.
- **Type VIII - Decision manipulation**: In this type of attack the final result is manipulated by the attacker.

Fingerprint scanners are divided into two major categories: optical and solid state sensors [6]. Beside optical sensors, the most common solid state sensors are capacitive scanners, thermal and ultrasound scanners [7]:

- **Optical scanners** take a visual image of the fingerprint using a digital camera while the finger is placed on a transparent prism.
- **Capacitive scanners** create an image by means of a small electric charge generated by miniature built-in capacitors that can store electricity. When the finger touches the scanner, it discharges these capacitors. Greater contact (ridges) causes more discharge; gaps between the skin and the sensor (valleys) cause less. The scanner measures the difference and determines the pattern.
- **Ultrasound scanners** use an ultrasound signal instead of light, and record the echo generated by ridges and valleys (as with light reflection, ridges and valleys have different echoes). This type of scanner does not need to be in contact with the finger, so it can be located under the screen.
- In **Thermal scanners** the finger is placed on pyroelectric material, which converts variations in temperature into voltage. The contact of the ridges with the sensing material causes a change in temperature, while the temperature remains constant under the valleys that are not in direct contact with the material.

In this paper we focus on Type I. attacks (direct attacks) against optical and capacitive fingerprint sensors which are the most common in mobile devices. We execute direct attacks by presenting the biometric characteristics of a registered user to the sensing device to access the system as an authorized user. We propose two methods (cooperative and non-cooperative) that can be used to bypass fingerprint sensors using a direct attack.

We note that there already exist a number of surveys on biometric authentication bypasses, however most of them are not precisely documented or the achieved success ratio is less than 80%-85%. We are not aware of any security research team who achieved 95% accuracy of bypassing fingerprint sensors using a direct attack. The main contributions of this paper can be summarized as follows:

1) Introducing a cooperative direct attack methodology where we were able to achieve 100% success ratio against optical and 96% success ratio against capacitive sensors (Section 3.1).
2) Introducing a non-cooperative direct attack methodology where we were able to achieve 98% success ratio against optical sensors (Section 3.2).
3) All methods have been precisely described in order to allow the research community to reproduce our results and to spur further research in this area (Section 3).

II. Related Work

Attacking fingerprint sensors has a long history. In 2002 Matsamuto et al [8] already presented that gummy fingers, namely artificial fingers that are easily made of cheap and readily available gelatin, were accepted with extremely high rates (68%-100%) by different fingerprint devices with optical or capacitive sensors. We note that the techniques descibed by Matsamuto are not effective against modern fingerprint sensors. Zhang Rui and Zheng Yan [9] classified and thoroughly reviewed the existing biometric authentication systems by focusing on the security and privacy solutions of them. They also claim that existing systems could be vulnerable to attacks such as replay attacks and suffer from user privacy intrusion. Galbally et all [10] conducted direct attack experiments using a highly competitive ISO minutia-based matcher and fake fingers generated from ISO/IEC 19794-2 templates printed on plastic cards as 2D barcodes. Jan Krissler known as ”Starbug” used the VeriFinger software [12] to demonstrate his method for faking fingerprints using only a few high-definition photographs of his target, German defence minister Ursula von der Leyen in 2014.

Recently a research team from Cisco's Talos security group [11] claimed an $\approx 80\%$ success rate using fake fingerprints and cooperative methods. During the experiments the researchers were not able to bypass encrypted pendrives such as Lexar Jumpdrive Fingerprint F35, and were not able to bypass the Windows Hello framework algorithm.

Mastering the Matsamuto method, and the method that Cisco's Talos security group used we are presenting two methodologies (cooperative and non-cooperative ones) where we were able to achieve almost perfect success ratio against the Windows Hello Framework, and against certified high security standard devices such as Lexar Jumpdrive Fingerprint F35 encrypted USB drive to bypass authentication and gain access to sensitive files.

All experiments were conducted in xen1thLabs Security Research Laboratory in Abu Dhabi, UAE. The methods presented in this paper are documented in detail and can be done without any expensive equipment

III. Methodology

There are many different factors when measuring the performance of a biometric system, however one of the most important are the False Acceptance Rates (FAR) and the False Reject Rates (FRR). False Acceptance Rate is the possibility of identifying an impostor as a legitimate user. False Rejection Rate is the possibility of identifying a legitimate user as an impostor. In a biometric system what one would like to achieve is a balance between the two error types, referred to as an

equal error rate (EER) [13]. EER refers to the point when the proportion of false acceptance is equal to the proportion of false rejection. Generally, the lower the equal error rate, the higher the accuracy of a biometric system is. If we represent FAR and FRR on a graph, EER can be viewed as the point where the two lines intersect [13].

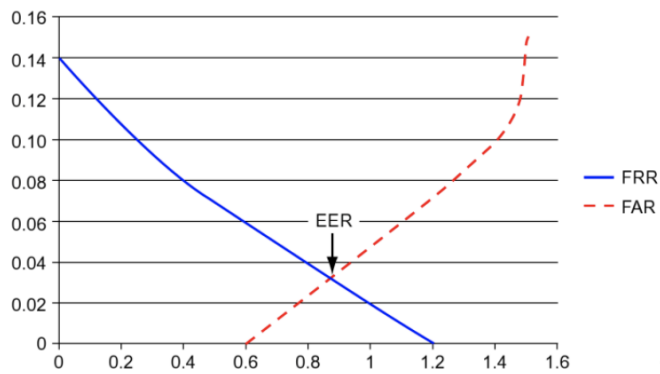

Fig. 2: Equal Error Rate

In a direct attack we would like to maximize the False Acceptance Rate (FAR), in such a way that a cloned fingerprint is always accepted.

A. Cooperative method

In a cooperative method, the owner of the fingerprint is involved in the process of creation of the fingerprint which is used to bypass the biometric sensor. The exact methodology can be described using the following methods:

First the mold of the finger of a legitimate user is created using ethylene-vinyl acetate EVA $(C_2H_4)_n(C_4H_6O_2)_m$ commonly known as "hot glue".

1) **Glue Gun Heat up:** Heat up the glue. We used a 65W hot glue gun (HL-603) to heat up the glue which was used to create the mold of the user (Figure 3). On Amazon that cost us less than $9.00. Alternatively, you could use a cigarette lighter ($0.50) to heat the glue stick which would bring the cost down further.
2) **Spreading:** After the glue is hot enough spread it on a piece of paper. We found that creating a boundary first and filling it in with the glue creates an even distribution and generally a flat surface to place your finger.
3) **Waiting:** Before creating the mold wait 25-35 secs to cool off the glue. This is important. If you try to do it earlier, there is a risk of serious skin burn and also destroying the cast.
4) **Creating the mold:** Apply some lubricant (we used petroleum jelly) to your finger first. This provides some heat insulation to the finger as well as avoids your finger from sticking to the mold. Slowly press your finger into the glue and gently lift your finger after 2-3 secs. Avoid exerting too much pressure as this might damage the mold making process. If done right, you should be able to see all ridges and valleys of the finger in the mold (Figure 4).
5) **Creating the final cast:** Put the PVA - $C_4H_6O_2$ (white glue or school glue (Figure 5)) into the mold. Spread it evenly using the tip of a piece of paper. The glue layer should be a maximum of about 1 millimeters in thickness (Figure 6). This is important as a thick layer will dry off unevenly and leave areas of glue that is still runny.
6) **Drying:** Using a hot air gun or an incandescent bulb dry the layer of glue off. This takes anywhere between 3 minutes to 5 minutes if the glue was applied evenly and thinly. Repeat the application and drying off procedure 3 times until a structurally sound cast has been created.
7) **Lifting the cast:** Using a sharp object gently peel off the sides of the cast. Be careful not to destroy the actual fingerprint impression (Figure 7).

From our experiments we found out that using EVA $(C_2H_4)_n(C_4H_6O_2)_m$ (hot glue) in conjunction with PVA - $C_4H_6O_2$ (white glue) makes the moulding and casting process economical, less error prone and durable.

Fig. 3: HL-603 glue gun

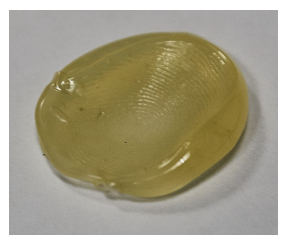

Fig. 4: Mold created using EVA hot glue

Fig. 5: White glue

Fig. 6: Mold filled with white glue

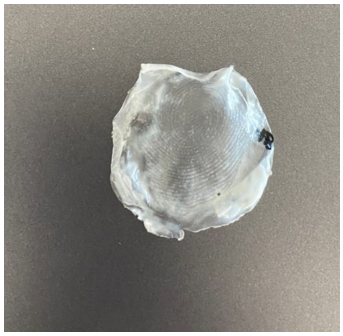

Fig. 7: Artificial fingerprint

B. Non-Cooperative method

In the non-cooperative method the latent fingerprints are obtained without the consent of the user. Also recreating a usable fingerprint mold with this type of attack is not a trivial process.

1) **Preparing the print for removal:** We carefully dust the latent fingerprint left behind by the victim with graphite powder and a soft bristled makeup brush. In our experiment we took the fingerprint from a notebook screen (Figure 8).
   The graphite powder was obtained from a 2HB pencil by scraping away at the lead using an X-ACTO knife. Alternatively, we could buy graphite powder commercially sold as a lubricant for door hinges. The dusting process should be carried out very lightly avoiding too much pressure when using the brush to avoid ruining the print. The idea is that the graphite powder should adhere to the oils present in the fingerprint and make it more visible. Any excess powder should be blown off using a Rubber Hand Pump Air Blower.
2) **Lifting the fingerprint:** We lift the now dusted fingerprint using stick tape (Figure 9), either scotch or transparent Sellotape. Gently rub the tape down over the fingerprint until you are confident that you have gone over the entire print.
   Next, peel the tape off in one slow controlled motion. Place the now lifted print sticky side down onto a sheet of A4 paper.
3) **Scanning and Digitization:** We then scan the lifted fingerprint and transfer it to a computer using a commercially available paper scanner. We found that a resolution of 1200DPI and above gave the best results (Figure 10) .
4) **Digitally enhancing the fingerprint:** The image is then digitally enhanced using open source image editing software to get the best ”noise free” negative horizontally flipped image with black background (Figure 11).
5) **Printing to transparencies:** The final “cleaned” image is then printed onto a transparency paper using a high resolution laser printer (Figure 12). We found that true 1200 DPI resolution laser printers gave the best result. It is very important to ensure that the final image size is exactly the same size as the actual finger print size. In order to achieve this we drew a bounding box on the A4 paper to act as markers to help align the printed transparency image with the actual image.
6) **Transfer image to PCB substrate:** The image was then transferred to a photosensitive Printed Circuit Board that has a copper cladding layer of 1oz (Figure 13). The exposure box used Three UV (350 450µm) FL-15 tubes to ensure high intensity ultra-violet radiation. The board was exposed for a total of 130 seconds.
7) **Developing the PCB:** The exposed image (Figure 14) was then developed using a developing solution made of sodium metasilicate Na2SiO3. Approximately, 50g of Na2SiO was mixed with 1 litre of 25° C water.
8) **Etching the PCB:** The PCB is etched using and Sodium Persulphate Na2S2O8 that was heated to 50° C to speed up the etching process (Figure 15,Figure 16).
9) **Creating the final cast:** The PCB is then coated with glue and dried off layer by layer.Each layer of white glue should be evenly applied and should be no more than 1 mm thick. The layer should then be dried off using a lamp or hot air. 3 layers later when the cast is structurally sound it is lifted off the board. The cast is now ready to be used to bypass optical sensors (Figure 17).

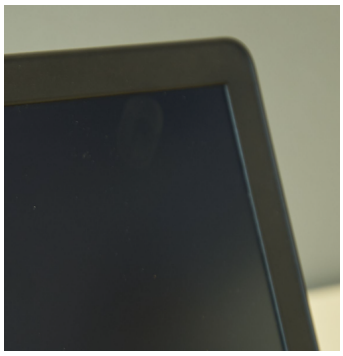

Fig. 8: Latent fingerprint on the screen

Fig. 9: Fingerprint lifted with graphite powder

Fig. 10: Scanned fingerprint

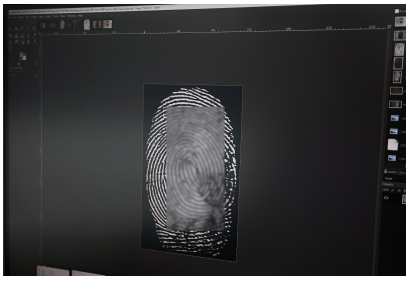

Fig. 11: Transformed fingerprint

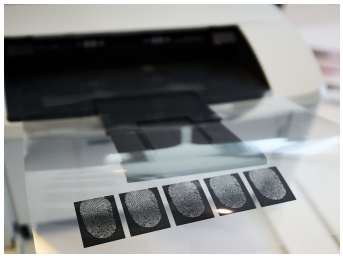

Fig. 12: Printing to transparency sheet

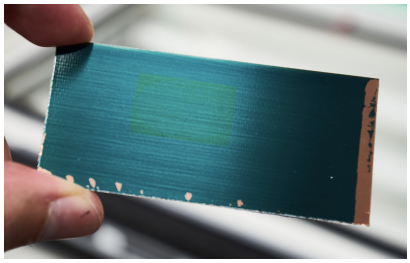

Fig. 13: PCB after UV exposure

Fig. 14: Developed PCB

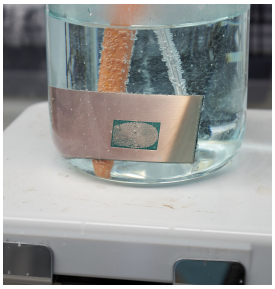

Fig. 15: Etching the PCB

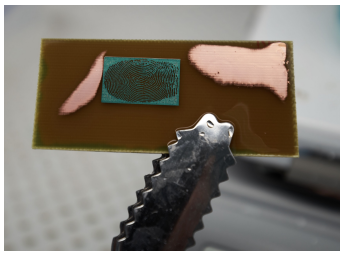

Fig. 16: Etched PCB

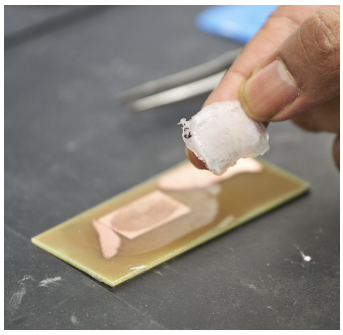

Fig. 17: Final cast

IV. Experimental results

Using the methodology presented in the previous section we were able to achieve almost 100% success rate in all cases.

In 2020 April Cisco's Talos threat intelligence and research group was able to achieve 95% success rate against a MacBook Pro, however they could not achieve even a single successful bypass on Windows 10 devices that use the Windows Hello framework and were not able to bypass authentication on encrypted USB drives from Verbatim and Lexar.

We were able to achieve 100% success rate on the Lexar Jumpdrive Fingerprint F35 USB encrypted drive and 96% success rate on a Windows Hello Framework (ASUS Zenbook UX461F Z14). The percentage is calculated based on 50 attempts for each device with the best fake fingerprint we were able to create. We would like to note a very important result in this research. When we created the first fake fingerprint we were able to achieve only 10%-17% success rate on an ASUS Zenbook UX461F Z14 Windows laptop. During the research it has turned out that the problem was not in the created fingerprint, rather the technique with which we used the fake fingerprint. After practicing for 10-15 minutes on the ASUS Zenbook Windows machine we were able to achieve 96% success rate (48 out of 50) with the same fake fingerprint and 100% success rate (50 out of 50) on the Lexar Jumpdrive Fingerprint F35 USB encrypted drive. In order to achieve better performance, and smooth authentication some devices are using machine learning techniques to adapt the fingerprint templates after every successful login attempt. In order to exclude this possibility we repeated the experiments with a new registered fingerprint, and we were able to achieve the same success ratio (without training). We can conclude that different devices require different techniques to bypass the fingerprint authentication. Precise pressure, appropriate angle, accurate "placement" of the fake fingerprint. This new result lead us to create the following methodology to bypass fingerprint sensors:

- **Create a test mold and test fake fingerprint**
- **Practice on the same test device**
- **After successful training (95%-100% success rate) one can start fooling the target device.**

During the experiments different types of devices were tested (see TABLE I) using the cooperative method and almost 100% success ratio was achieved using the above mentioned methodology in every single case. We note that Suprema Biostation 2 door access control is a high end security product, the only optical sensor we have tested. In the experiments we achieved 100% success rate on this device using the cooperative and non-cooperative method as well. We encourage the research community to try our non-cooperative method on different optical sensors.

TABLE I: Tested devices

Device name	Sensor Type	Success rate
2017 MacBook Pro (High Sierra)	capacitive	100%
2019 Macbook Air (Big sur)	capacitive	100%
ASUS Zenbook UX461F Z14	capacitive	96%
HUAWEI P20	capacitive	100%
HUAWEI P9	capacitive	100%
iPhone 8 Plus	capacitive	100%
Lexar Jumpdrive F35	capacitive	100%
Suprema Biostation 2 (door accces)	optical	100%

V. Conclusion and further research

We have presented a cooperative methodology to bypass optical and capacitive fingerprint sensors with very high accuracy. We also presented a non-cooperative methodology for bypassing optical sensors. However, we are at work in trying to find a way to bypass capacitive sensors using non cooperative techniques using a minimalist approach. Currently, the problems we face are in determining the actual height map along the Z axis in 3D space to spoof the capacitive sensors. This is really challenging to derive from a 2D image without using a 3D printer.

Acknowledgment

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